The empirical recurrence for OEIS A296648: recovering a conjecture that is not written down

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Abstract

OEIS A296648 records "Empirical recurrence of order 58", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 107$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 58, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 58 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A296648 is "Number of nX5 0..1 arrays with each 1 adjacent to 0, 1 or 3 king-move neighboring 1s.". It has offset 1 and begins

$$24, 248, 1985, 20782, 212537, 2022756, 20337282, 202790850, \ \ldots$$

The entry states:

> Empirical recurrence of order 58 (see link above)

As of the "Last modified" line on the live entry (Dec 17 2017 15:46:28, revision 4) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1056$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 107$ of the 1056, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' =$ 107.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 214$ exact terms of the model returns order 58 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{58} c_i\, a(n-i),$$

c_1	4	c_{16}	−596698765	c_{31}	35652542015	c_{46}	16647851
c_2	36	c_{17}	−2433858766	c_{32}	86585379830	c_{47}	10199161648
c_3	420	c_{18}	−2375829129	c_{33}	68472631743	c_{48}	5144791138
c_4	−770	c_{19}	4060344303	c_{34}	12303304499	c_{49}	−1306583988
c_5	−7960	c_{20}	9367885243	c_{35}	−76548949675	c_{50}	−853387852
c_6	−47734	c_{21}	2018742447	c_{36}	−100737564030	c_{51}	−163135700
c_7	50083	c_{22}	−13019233718	c_{37}	−55874389113	c_{52}	−230177896
c_8	633739	c_{23}	−14943295382	c_{38}	18169165030	c_{53}	129640048
c_9	2226709	c_{24}	6056976497	c_{39}	59713802342	c_{54}	9363744
c_{10}	−1688106	c_{25}	27331811913	c_{40}	55756541600	c_{55}	−54592320
c_{11}	−22081199	c_{26}	17237750829	c_{41}	23078246726	c_{56}	352640
c_{12}	−47832573	c_{27}	−19334992177	c_{42}	−7755245396	c_{57}	911872
c_{13}	42148192	c_{28}	−48179506344	c_{43}	−23353444923	c_{58}	−258048
c_{14}	334693621	c_{29}	−38055563525	c_{44}	−16288121222		
c_{15}	506780571	c_{30}	−1911401346	c_{45}	−12294254993		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 58, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $58+4$ terms removed returns order 58 as well, so $\deg q_{\min} = 58 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $58+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 58 that this sequence satisfies — and that family turns out to have exactly one member.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A296648.

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